

18.06 Linear Algebra - Lecture 10-12

Textbook Problem Sets

• Problem Set 3.5 pg 190

1) a) 7×9 matrix rank $\leq \infty$ $r=5$ ✓

$$n \times m = 7 \times 9$$

$$\dim(C(A)) = \dim(C(A^T)) = r = 5$$
 ✓

$$\dim(N(A)) = n - r = 7 - 5 = 2$$
 ✓

$$\dim(N(A^T)) = m - r = 9 - 5 = 4$$
 ✓

$$\text{Sum dims} = (5) + (5) + (2) + (4) = 16 \quad \{(m+n) = (7+9) = (16)\}$$

b) 3×4 matrix $r=3$

$$C(A) = \mathbb{R}^3$$

$$\dim(N(A^T)) = m - r = 3 - 3 = 0 \rightarrow$$

$$N(A^T) = \{0\}$$

3) $A = \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 2 & 4 & 6 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$$\Rightarrow R = \begin{bmatrix} 0 & 1 & 2 & 0 & -2 \\ 0 & 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$r=2$$

$$A \text{ size } m \times n = 3 \times 5$$

$$C(A): \text{basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} \right\}$$
 ✓

$$R_x = 0$$

$$C(A^T): \text{basis} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$
 ✓

$$\text{Set } x_1 = 1, x_3 = 0, x_5 = 0$$

$$x_2 + 2(1) - 2(0) = 0 \Rightarrow x_2 = -2$$

$$x_2 = 0$$

$$x_4 + (0) = 0 \Rightarrow x_4 = 0$$

$$\text{Set } x_3 = 1, x_1 = 0, x_5 = 0$$

$$x_2 + 2(1) - 2(0) = 0 \Rightarrow x_2 = -2$$

$$x_4 + (0) = 0 \Rightarrow x_4 = 0$$

$$s_2 = \begin{bmatrix} 0 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$s_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Set } x_5 = 1, x_1, x_3 = 0: x_2 + 2(0) - 2(1) = 0 \Rightarrow x_2 = 2$$

$$x_4 + 2(1) = 0 \Rightarrow x_4 = -2$$

$$\therefore x_4 = -2$$

$$s_3 = \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix}$$

$$N(A) \text{ basis} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$
 ✓

3. con.)

$$[A | I_m] \rightarrow [R | E_m]$$

$$\left[\begin{array}{ccccc|ccc} 0 & 1 & 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & 1 & 2 & 4 & 6 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right] \Rightarrow \left[\begin{array}{ccccc|ccc} 0 & 1 & 2 & 3 & 4 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \end{array} \right]$$

$$N(A^T) = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\} \checkmark$$

7) 3×3 matrix $A \rightarrow$ invertible

$$r = 3 \quad n = m = 3$$

3×6 matrix $B = [A \ A]$

$$n - r = 3 - 3 = 0$$

$$m - r = 3 - 3 = 0$$

$$\text{basis } C(A) = \mathbb{R}^3 \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ basis } N(A) = \{ \mathbf{0} \} \checkmark$$

$$\text{basis } C(A^T) = \mathbb{R}^3 \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\} \text{ basis } N(A^T) = \{ \mathbf{0} \} \checkmark$$

Yes that is the column space but the basis vectors are $\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$

11) $A \rightarrow m \times n$ r some b have no solution for $Ax = b$.

a) $r < m$; $r < n$ ✓

b) $r < m$ which means there ~~are~~ ^{is} ~~free variables~~ more than just the zero vector ~~within~~ within $N(A^T)$. ✓

$\hookrightarrow m - r > 0 \rightarrow$ Better representation.

14) $A = \begin{bmatrix} \quad \end{bmatrix} \begin{bmatrix} \quad \end{bmatrix} = A_1 A_2$

$$A_1 \text{ rank} = 3 = A_2 \text{ rank}$$

$$\therefore A \text{ rank} = 3$$

$$A_{m \times n} = A_{1(3 \times 3)} \times A_{2(3 \times 4)} = A_{(3 \times 4)} \quad m = 3 ; n = 4$$

$$\text{basis } C(A) = \left\{ \begin{bmatrix} \quad \end{bmatrix} \right\}$$

$$\dim C(A) = \dim C(A^T) = r = 3$$

$$\dim N(A) = n - r = 4 - 3 = 1$$

$$\dim N(A^T) = m - r = 3 - 3 = 0$$

$$17) A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$r = 2$$

$$\dim C(A) = 2 \quad \dim N(A) = 1$$

$$\dim C(A^T) = 2 \quad \dim N(A^T) = 1$$

$$\text{basis } C(A^T) = \{ (0, 1, 0), (0, 0, 1) \}$$

$$\text{basis } C(A) = \{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \}$$

$$\text{basis } C(A) = \{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \}$$

$$Ax = 0$$

$$\text{Set } x_1 = 1$$

$$\therefore 0 + x_2 + 0 = 0 \quad \therefore x_2 = 0$$

$$x_3 = 0 \quad \therefore x_3 = 0$$

$$x = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \Rightarrow \text{basis } N(A^T) = \{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \}$$

$$C(A) \Rightarrow \text{span } xy \text{ plane}$$

$$C(A^T) \Rightarrow \text{span } yz \text{ plane}$$

$$N(A) \Rightarrow \text{span } x \text{ axis}$$

$$N(A^T) \Rightarrow \text{span } z \text{ axis.}$$

$$20) \text{ } x = \{ (-1, 2, 0, 0), (-1/4, 0, 3, 1) \}$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 4 & 1 \end{bmatrix} \begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = ER$$

$$(3 \times 3) \cdot (3 \times 4) \rightarrow (3 \times 4)$$

$$\text{rank} = 2$$

$$Rx = 0$$

$$\begin{bmatrix} 4 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 & 0 & 1/4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\text{Set } x_2 = 1, x_4 = 0$$

$$x_1 + 1/2(1) + 0 + 1/4(0) = 0 \quad x_1 = -1/2$$

$$x_3 + 3(0) = 0 \quad \therefore x_3 = 0$$

$$x_{s1} = \begin{bmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Set } x_4 = 1, x_2 = 0$$

$$x_1 + 1/2(0) + 0 + 1/4(1) = 0 \quad \therefore x_1 = -1/4$$

$$x_3 + 3(1) = 0 \quad \therefore x_3 = -3$$

$$x_{s2} = \begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix}$$

$$N(A) \text{ basis} = \left\{ \begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 1 \\ 0 \\ 0 \end{bmatrix} \right\}$$

Perpendicularity Check: $x_1, x_2 = 0 \Rightarrow$ perpendicular

$$\left\{ \begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix}, \begin{bmatrix} -1/2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$\Rightarrow \begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix}_{4 \times 1} \cdot \begin{bmatrix} 4 & 2 & 0 & 1 \end{bmatrix}_{1 \times 4} =$$

$$\begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix} \cdot (4 \ 2 \ 0 \ 1) = (-1/4)(4) + (0)(2) + (-3)(0) + (1)(1) = -1 + 1 = 0 \rightarrow$$

$$\begin{bmatrix} -1/4 \\ 0 \\ -3 \\ 1 \end{bmatrix} \cdot (0 \ 0 \ 1 \ 3) = (-1/4)(0) + (0)(0) + (-3)(1) + (1)(3) = -3 + 3 = 0 \rightarrow$$

$$\begin{bmatrix} -1/2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot (4 \ 2 \ 0 \ 1) = (-1/2)(4) + (0)(2) + 0 + 0 = -2 + 2 = 0 \rightarrow$$

$$\begin{bmatrix} -1/2 \\ 0 \\ 0 \\ 0 \end{bmatrix} \cdot (0 \ 0 \ 1 \ 3) = (-1/2)(0) + (0)(0) + (0)(1) + (0)(3) = 0 \rightarrow$$

$\therefore$ Solutions to $Ax=0$ (Nullspace) is perpendicular to rows of A .

b) $A^T y = 0$ ~~done~~

$$\dim N(A^T) = m - r = 3 - 2 = 1 \text{ solution to } A^T y = 0$$

$$23) A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix}$$

(3x2) (2x3) $\rightarrow$ (3x3)

$$A_{(m \times n)} = A_{(3 \times 3)}$$

$$\begin{bmatrix} 3 & 0 & 3 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \text{ rank} = 2 = r$$

$$\therefore \text{basis } C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}_x = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 7 \end{bmatrix} \right\}$$

$$\text{basis } C(A^T) = \{ (3 \ 0 \ 3); (1 \ 1 \ 2) \} \checkmark$$

$\rightarrow$ Specifically $\circ$ rank (3,2) x (2,3) $\leq$ either so rank ≤ 2
 $\circ$ Rank 2 for (3,3) $\circ$ Not invertible

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Textbook Problem Sets

• Problem Set 10.1 pg 459

1) $A_{3 \times 3} =$ ~~$\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$~~ ~~$\begin{bmatrix} +1 & -1 & 0 \\ 0 & +1 & -1 \\ 0 & 0 & +1 \end{bmatrix}$~~ $\begin{bmatrix} +1 & -1 & 0 \\ +1 & 0 & -1 \\ 0 & +1 & -1 \end{bmatrix}$ ~~*~~

Incidence Matrix $\rightarrow$ rows are edges
 $\rightarrow$ cols are nodes.

$$A_{3 \times 3} = \begin{bmatrix} -1 & +1 & 0 \\ -1 & 0 & +1 \\ 0 & -1 & +1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$N(A) = \mathbf{0} \times \begin{bmatrix} -1 & +1 & 0 \\ -1 & 0 & +1 \\ 0 & -1 & +1 \end{bmatrix} = \begin{bmatrix} -1 & 0 & +1 \\ -1 & +1 & 0 \\ 0 & -1 & +1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \text{ rank 2}$$

Set $x_3 = 1$

$$x_2 - x_3 = 0$$

$$x_2 - (1) = 0 \quad x_2 = 1$$

$$x_1 - (1) = 0 \quad x_1 = 1$$

$$\therefore x_s = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad x_s = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$\hookrightarrow$ Every connected graph has $x_n = \{1\}$ (same potential)

3) $A = \left[\begin{array}{ccc|c} -1 & +1 & 0 & b_1 \\ -1 & 0 & +1 & b_2 \\ 0 & -1 & +1 & b_3 \end{array} \right]$

$$U = \left[\begin{array}{ccc|c} -1 & 0 & 1 & b_2 \\ -1 & 1 & 0 & b_1 \\ 0 & -1 & 1 & b_3 \end{array} \right] = \left[\begin{array}{ccc|c} -1 & 0 & 1 & b_2 \\ 0 & 1 & -1 & b_1 - b_2 \\ 0 & -1 & 1 & b_3 \end{array} \right] = \left[\begin{array}{ccc|c} -1 & 0 & 1 & b_2 \\ 0 & 1 & -1 & b_1 - b_2 \\ 0 & 0 & 0 & b_3 + b_1 - b_2 \end{array} \right]$$

$\hookrightarrow$ (Non-zero rows are edges 1 and 2.) $\rightarrow$ (3)? Microblat?
 (Zero row comes from the closed loop)

$$5) V = (f_1, f_2, f_3) \quad x = (1, 1, 1)$$

$$A^T y = f \rightarrow \text{Kirchhoff's current law.} \checkmark$$

$$A = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\text{rank } r = 2 \\ m = 3$$

$$\dim N(A^T) = m - r = 3 - 2 = 1$$

$$A^T = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \rightarrow$$

$$\begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -1 & -1 & 0 & | & f_1 \\ 0 & -1 & -1 & | & f_1 + f_2 \\ 0 & 1 & 1 & | & f_3 \end{bmatrix} \Rightarrow \begin{bmatrix} -1 & -1 & 0 & | & f_1 \\ 0 & -1 & -1 & | & f_1 + f_2 \\ 0 & 0 & 0 & | & f_1 + f_2 + f_3 \end{bmatrix} \checkmark$$

$$f_1 + f_2 + f_3 = 0 \Rightarrow \text{Solvable}$$

$$(1) + (-1) + (0) = 0$$

$$\circ \circ f_s = (1, -1, 0) \checkmark$$

$$(1) + (1) + (1) \neq 0$$

$$\circ \circ f_u = (1, 1, 1) \checkmark$$

7) Conductances $C_1 = 1$ $C_2 = C_3 = 2$
 $A^T C A$

$$A^T = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \quad A = \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$A^T C A = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -3 \\ -1 \\ 4 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} =$$

(3x3) (3x1) (3x3) (3x1) (3x3)

$$A^T C A = \begin{bmatrix} -1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -1 & -2 & 0 \\ 1 & 0 & -2 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 3 & 1 & -2 \\ -1 & 3 & -2 \\ -2 & -2 & 4 \end{bmatrix} \times \begin{bmatrix} 3 & 1 & -2 \\ -1 & 3 & -2 \\ -2 & -2 & 4 \end{bmatrix}$$

$A^T C A x = f \quad (A x = b)$

$$\left[\begin{array}{ccc|c} 3 & 1 & -2 & 1 \\ -1 & 3 & -2 & 0 \\ -2 & -2 & 4 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -2 & 1 \\ 0 & 10 & -8 & 3 \\ -2 & -2 & 4 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -2 & 1 \\ 0 & 5 & -4 & 1/2 \\ 0 & -4 & 8 & -1 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -2 & 1 \\ 0 & 5 & -4 & 1/2 \\ 0 & 0 & 24 & -3 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & 1 & -2 & 1 \\ 0 & 5 & -4 & 1/2 \\ 0 & 0 & 8 & -1 \end{array} \right]$$

$$5(-4) + 4(5) = 0$$

$$5(8) + 4(-4) = 24$$

$$40 - 16$$

$$5(-1) + 4(1/2) = -3$$

$$-5 + 2$$

$$3(-2) + 2(1)$$

$$3(4) + 2(-2)$$

$$12 - 4$$

$$3(-1) + 2(1)$$

$$\left[\begin{array}{ccc|c} 3 & -1 & -2 & 1 \\ 0 & 8 & -8 & 0 \\ -2 & -2 & 4 & -1 \end{array} \right] \Rightarrow \left[\begin{array}{ccc|c} 3 & -1 & -2 & 1 \\ 0 & 8 & -8 & 0 \\ -2 & -2 & 4 & -1 \end{array} \right]$$

$$3(-2) + 2(3) = 0$$

$$3(-2) + 2(1) =$$

$$-6 - 2 = -8$$

$$3(4) + 2(-2)$$

$$12 - 4 = 8$$

$$3(-1) + 2(1) = -1$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & -1 & -2 & 1 \\ 0 & 8 & -8 & 0 \\ 0 & 0 & 0 & -1 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & -1 & -2 & 1 \\ 0 & 8 & -8 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$\Rightarrow \left[\begin{array}{ccc|c} 3 & -1 & -2 & 1 \\ 0 & 8/3 & -8/3 & 1/3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Set $x_3 = 0$

$$x_p = \begin{bmatrix} 3/8 \\ 1/8 \\ 0 \end{bmatrix}$$

$$0 - C A x = - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 3/8 \\ 1/8 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ -2 & 0 & 2 \\ 0 & -2 & 2 \end{bmatrix} \begin{bmatrix} 3/8 \\ 1/8 \\ 0 \end{bmatrix} = \begin{bmatrix} -1/4 \\ -3/4 \\ -1/4 \end{bmatrix} = \begin{bmatrix} 1/4 \\ 3/4 \\ 1/4 \end{bmatrix}$$

∴ basis $N(A^T) = \{ \mathbf{0} \}$ ✓

∴ basis $C(A^T) = \{ (1, 2, 3, 4) ; (0, 1, 2, 3) ; (0, 0, 1, 2) \}$ ✓

∴ ~~82~~ Special Solution

Set $x_4 = 1$ $x_3 + (1) = 0$ $x_3 = -1$

$$\begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

set $x_4 = 1$

$x_3 + 2(1) = 0 \therefore x_3 = -2$

$x_2 - 1(1) = 0 \therefore x_2 = 1$

$x_1 = 0$

$S_1 = \begin{bmatrix} 0 \\ 1 \\ -2 \\ 1 \end{bmatrix}$

∴ basis $N(A) = \left\{ \begin{bmatrix} 0 \\ 1 \\ -2 \\ 1 \end{bmatrix} \right\}$ ✓

∴ basis $C(A) = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} ; \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} ; \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} \right\}$ ✓ $\{ C(A) = \mathbb{R}^3 \}$

spans n dimension
use standard basis